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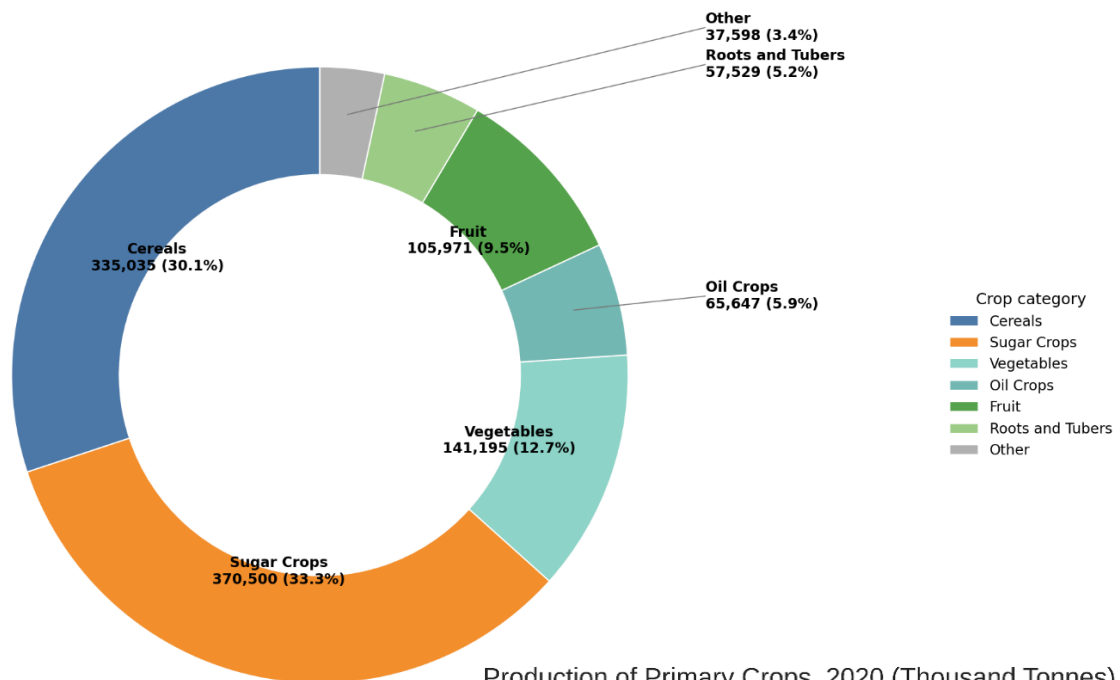
Understanding Human-Elephant Interactions in Agricultural Frontiers of India

1. Case Introduction

India harbours roughly 8% of the global biotic biodiversity and four of the global biodiversity hotspots[1]. These rich habitats provide housing to roughly 60% of the global population of the endangered Indian Elephant(*Elephas maximus indicus*)[2]. According to the latest DNA-based mark and recapture method, the largest population of the Asian elephants(*Elephas maximus*) is estimated at 22,500 individuals in India[3]. These majestic giants are a **National Heritage** species under Schedule I of the Wildlife Protection Act 1972[2]. With India's huge population, human-elephant conflict(HEC) is an inevitable, complex conservation issue. Conflict causes ~500 human and 100 elephant deaths annually, plus 1 million hectares of crop/property-loss affecting rural families[2][4]. This essay aims to peek at the problem through ecological, anthropogenic, and cultural lenses, evaluating the positions of the stakeholders, exploring tried and tested interventions, and possibly suggest some new regional, legal, technological, and community-based solutions.

2. Agricultural Context of India

Production of Primary Crops in India, 2020 (Thousand Tonnes)



Production of Primary Crops, 2020 (Thousand Tonnes)

Agriculture remains the backbone of the Indian rural society. Roughly 17% of the massive Indian GDP is currently drawn from agriculture, while 65% of the country's employment is still agrarian in nature[5][6]. Seen in parts of Figure-1, India is the second largest global producer of sugarcane, rice, wheat, cotton, and other crops[2]. The FAO statistics elaborate that cereals(especially rice and wheat), oil-crops(ground nut, soybean, sunflower, mustard, sesame, etc), and cotton(shown under “other”) dominate under the category of cultivated area in India as seen in Figure-2. However, in terms of production value, Figure-3 illustrates that while cereals retain the top position, vegetables, fruits and cotton seek the next highest values in

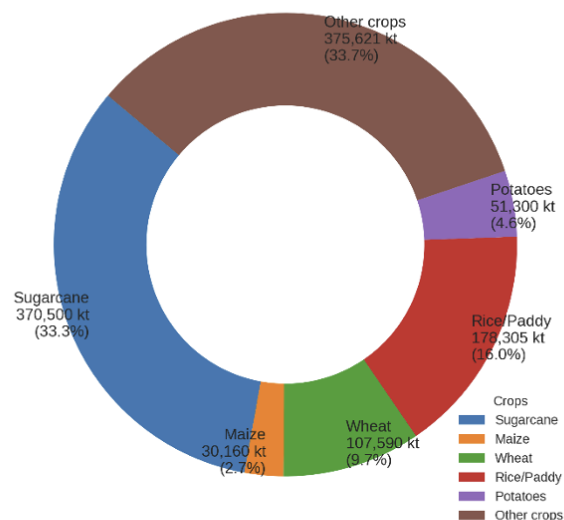
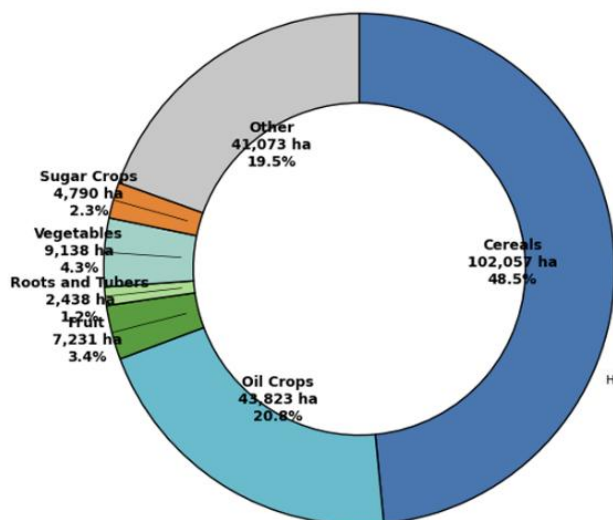


Figure 1: Pie charts showing production of primary crops and staple commodities by weight (kt) for India in 2020. Charts are based on data from FAO Statistics, 2022 [7].

the local and export markets. 86% of farmers own under one hectare, making each crop-rotation vital for livelihoods [6].

Harvested Area of Primary Crops in India, 2020 (Thousand Hectares)



Harvested Area of Primary Crops in UK, 2020 (Thousand Hectares)

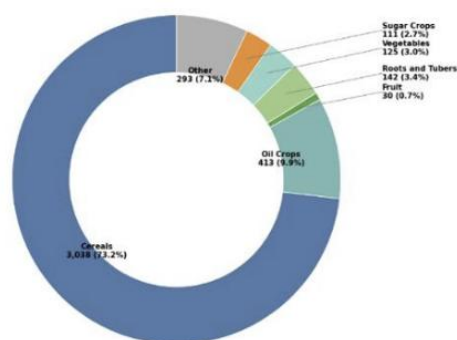
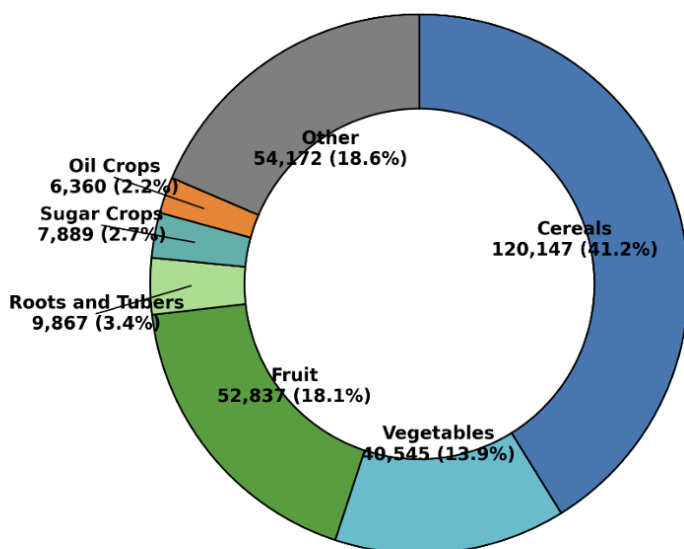


Figure 2: Pie charts showing harvested area of primary crops (million USD) in India and UK in 2020. Charts are based on data from FAO Statistics, 2022 [7].

Production Value of Primary Crops in India, 2020 (USD Million)



Production Value of Primary Crops in UK, 2020 (USD Million)

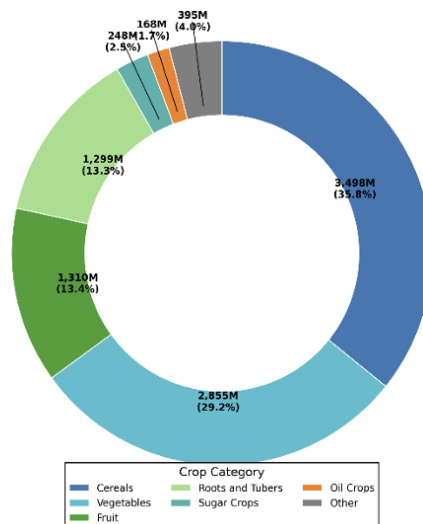
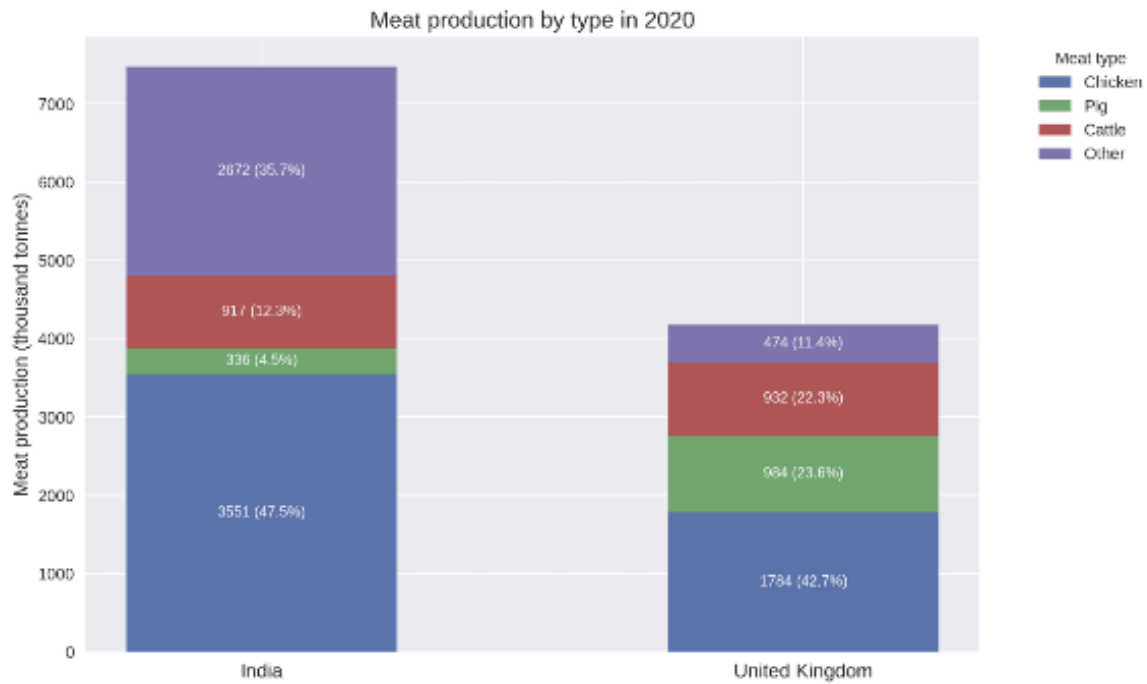


Figure 3: Pie charts showing production value of primary crops (million USD) in India and UK in 2020. Charts are based on data from FAO Statistics, 2022 [7].

Figure 4: Stacked bar chart comparing the production of various meat types by weight (kt) for India and UK in 2020. Charts are based on data from FAO Statistics, 2022 [7].



India mainly consumes chicken and goat, with smaller groups eating pork and buff meat(Figure-4). Being a land of cows, beef is banned throughout the nation, and almost all produced buff meat is exported. Resultantly, India is the largest milk and buff meat producer

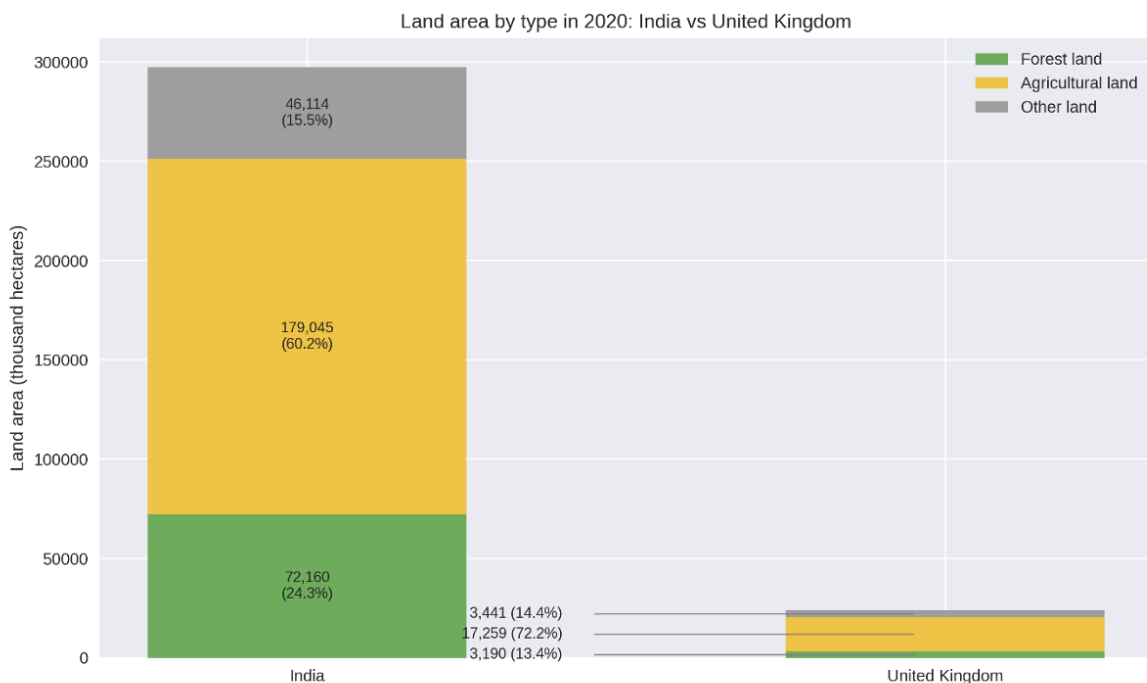


Figure 5: Stacked bar chart comparing the aggregated land area (kHa) by land-use types for India and UK in 2020. Charts are based on data from FAO Statistics, 2022 [7].

globally, accounting for about 24% of global milk production. It also holds the position as the second largest eggs and goat meat producer[7].

A smaller but equally valuable proportion of land is used for production forests, as seen in Figure-5. Timber includes Teak, Acacia, Eucalyptus, Poplar, Bamboo, used for furniture, paper, and crafts. As the country has a large wood market, the exports are minimal[8].

In short, India's agricultural mosaic of small-value, sugar-scented farms near forests sets the stage for the perfect big-footed animal heists.

3. Case Area Description and Conflict Premise

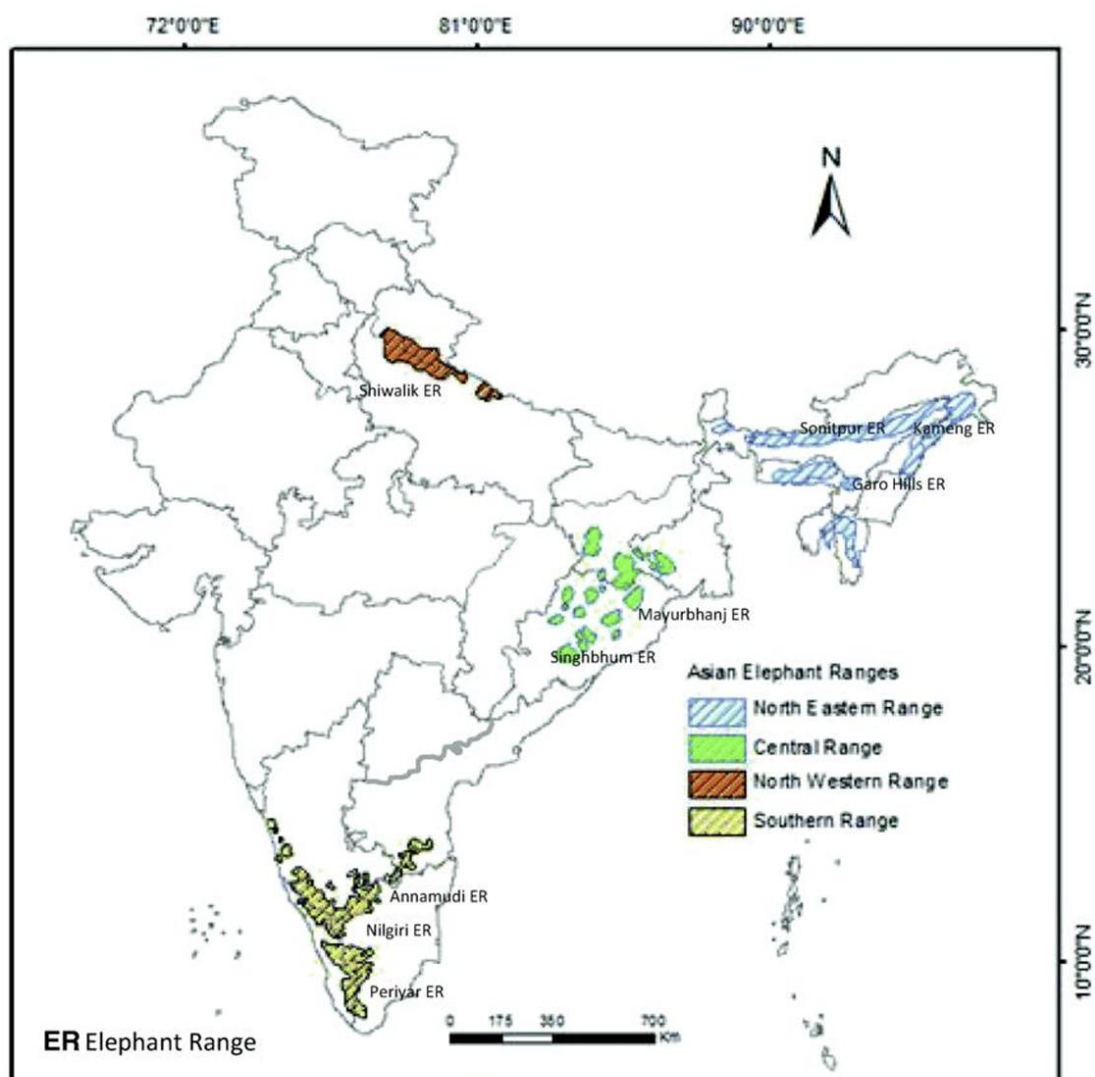


Figure 6: The map shows the distribution of the sub-populations of the Indian Elephant (*Elephas maximus indicus*) present in India in 2013. The figure is borrowed from Padalia et. al. (2019) [9].

Although it appears that there are four distinct elephant corridors in Figure-6, there are 5 genetically distinct sub-populations of Indian Elephants which rarely interact with each other: one in Uttarakhand(north-India), one which covers the central and north-eastern range, and three in the Western-Ghats(located in south-west India). These populations don't merge mainly due to geographical barriers. This also indicates a mechanism of how a conflict hotspot can shape elephant memory[10]. According to Figure-7, the southern genetics account for about 54%, about 38% is central-eastern, and the northern pub-population is about 4% of the total. In all, 150 elephant corridors have been identified as of 2023 which connect the various sub-populations together[11].

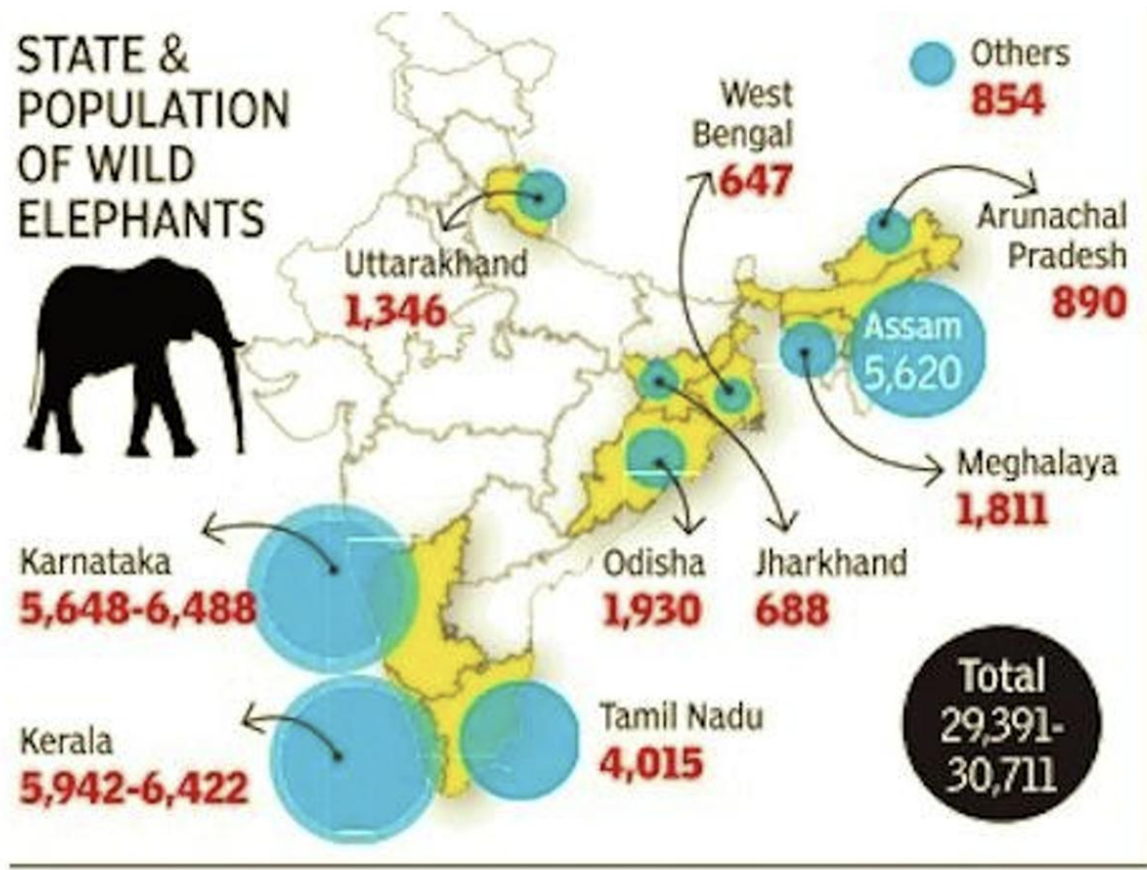


Figure 7: The map shows the state-wise population distribution of the Indian Elephant (*Elephas maximus indicus*) present in India between 2014-2016. The figure was published by "The Times of India" in 2017 using data from the Ministry of Environment, Forests and Climate Change (MoEFCC) [12].

When associated with the vegetation distribution in Figure-8, the elephant-human interface tends to be a diffuse and intermittent patchwork of farmlands and forest, joined by scrubland[4]. Minimal barriers let bulls or herds forage in crop-fields. Seasonality shapes crop-raiding intensity, creating new hotspots. Herds create new routes when corridors are blocked, forming fresh conflict hotspots[2][14].

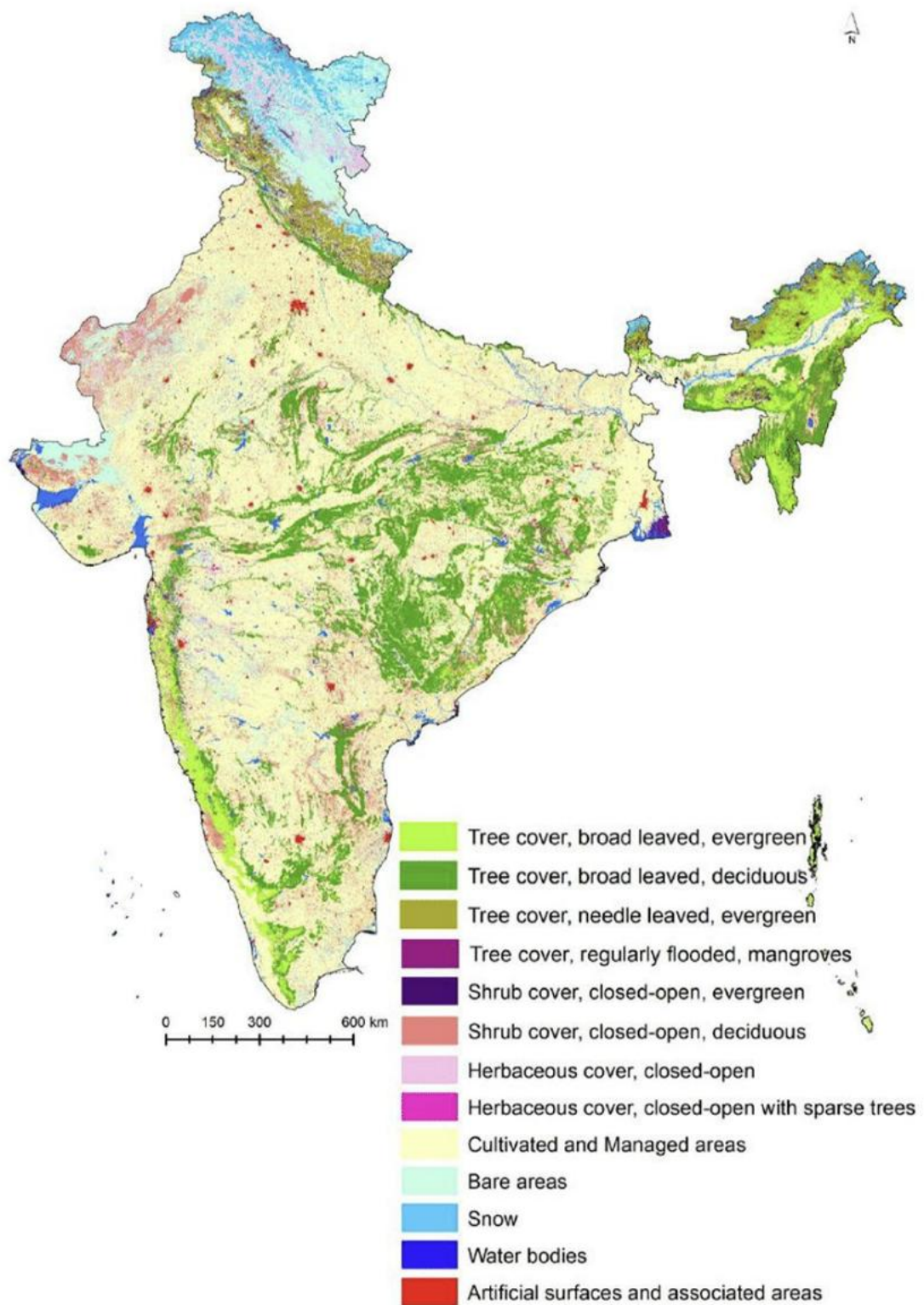


Figure 8: The map shows the land-use/land cover and natural vegetation distribution of India in 2005. The figure was sourced from Chintala et al (2015)[13].

The most worrisome result of HECs is ofcourse elephant and human fatalities. According to Figure-9, at least 9% of total elephant deaths are anthropogenic in nature. Though a small proportion, it can have lasting impacts on elephant ecology as well as their environment.

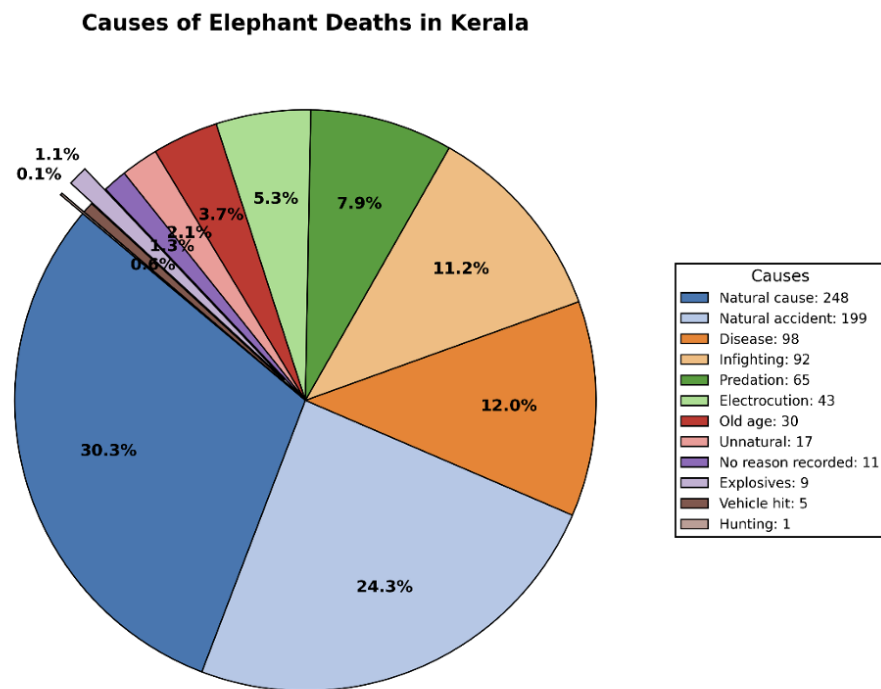


Figure 9: The pie chart shows the spread of wild elephant death causes in Kerala, India between 2023-2024. The figure is based on data published by “The Times of India” in Sept, 2025 [12].

4. Governance and Policy Setting

At the national level, India has listed the Asian Elephant under Schedule I(highest protection status) of the Wildlife Protection Act, 1972, which is a strong legal-framework wherein killing or generally capturing elephants is a serious offense[2]. The Indian Forest Act, 1927 regulates the protection of forests, which is the primary habitat for Asian Elephants[15]. Other schemes include Wildlife Habitat Development and the 2016-Afforestation Fund Act

Additionally, the National Elephant Conservation Authority(NECA) is the policy advisory to the Government of India. Under their guidance, the Ministry of Environment, Forest and Climate Change(MoEFCC) launched “Project Elephant” in 1992 to monitor elephant populations, fund protected areas, and compensate HEC affected communities[11].

Project Elephant created 33 reserves to conserve critical habitats. However, the reserves do not provide any legal protection themselves unless they also overlap with other protected areas or reserve forests[15].

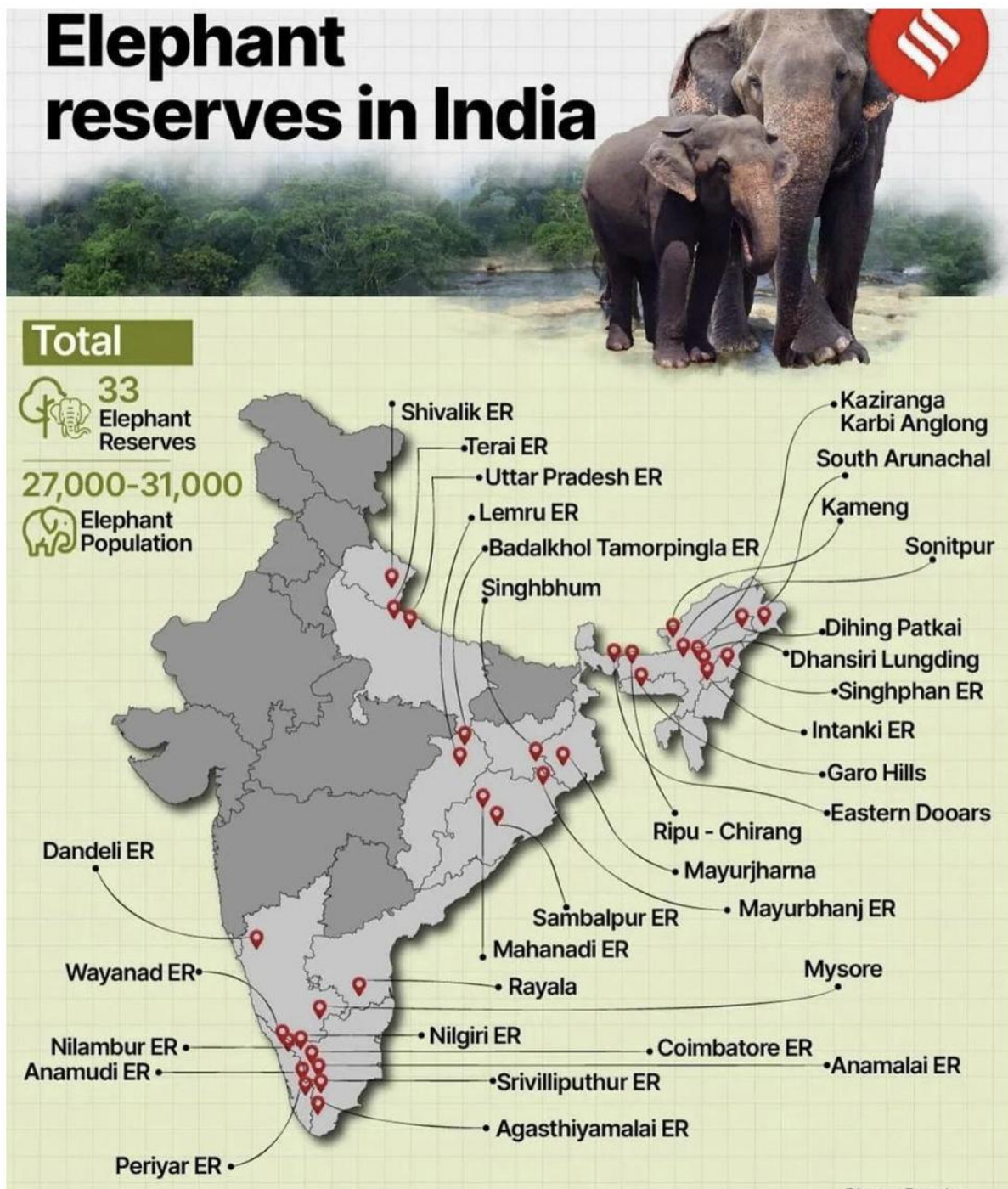


Figure 10: The map shows the distribution of the national elephant reserves present in India in 2013. Figure originally created using Pexel and published by "The Indian Express" [16].

In 2023, MoEFCC also identified 150 elephant corridors, which are now official blueprints for land-use planning and anti-infrastructure measures[11]. From an operations perspective, the Ministry has issued an HEC management guideline in 2022, which establishes SOPs and encourages HEC avoidance measures like trip alarms, community patrols, and deterrent crop-choice strategies. It has also established inter-departmental "hotspot" committees and rapid-response teams[15]. On the judicial side, the National Green Tribunal(NGT) offers quick resolution on environmental/conservation disputes[29].

Internationally, India signed CITES, banning elephant and ivory trade under Appendix I.

5. Cultural Setting

While not the primary subject of this essay, elephants are a cultural icon for India's cultural and religious fabric. The elephant-headed god "Ganesha" is one of the most widely worshipped deities in almost every Hindu household. Prayers to Ganesha precede new ventures, believed to bring luck and longevity(Figure-11).



Figure 11: The image shows the procession of the elephant-headed god "Ganesha" during the culmination of the 10-day long celebration of Ganesh Festival. The image is taken in Mumbai, India and is sourced from the "Picxy" stock-image website [17].

In South-India, captive elephants are adorned with ornaments in temples during the ceremonies and processions of Thrissur Pooram as they are considered the embodiment of Ganesha(Figure-12). Historically, kings owned elephants as symbols of power. Hence, the Indian Government has designated the elephant as the national heritage animal[2].



Figure 12: The image shows elephant worship and procession at a temple in Kerala, India. The image is sourced from X [18].

However, cultural tolerance can be a double-edged sword for HECs. Public sympathies make killing elephants taboo, despite past injuries. However, the stressful conditions cause diseases and mortality in captive temple elephants[19].

Some tribal communities fear elephants as forest spirits and create ‘Sacred Groves. In 2024, MoEFCC recommended a policy to designate these groves as “community reserves”[20]. HEC responses depend on local culture, religion, community, and politics.

6. HEC Mechanisms and Drivers

Following are the brief delineations for possible causes of HEC:

6.1 . Dangerous Agricultural Safeguards

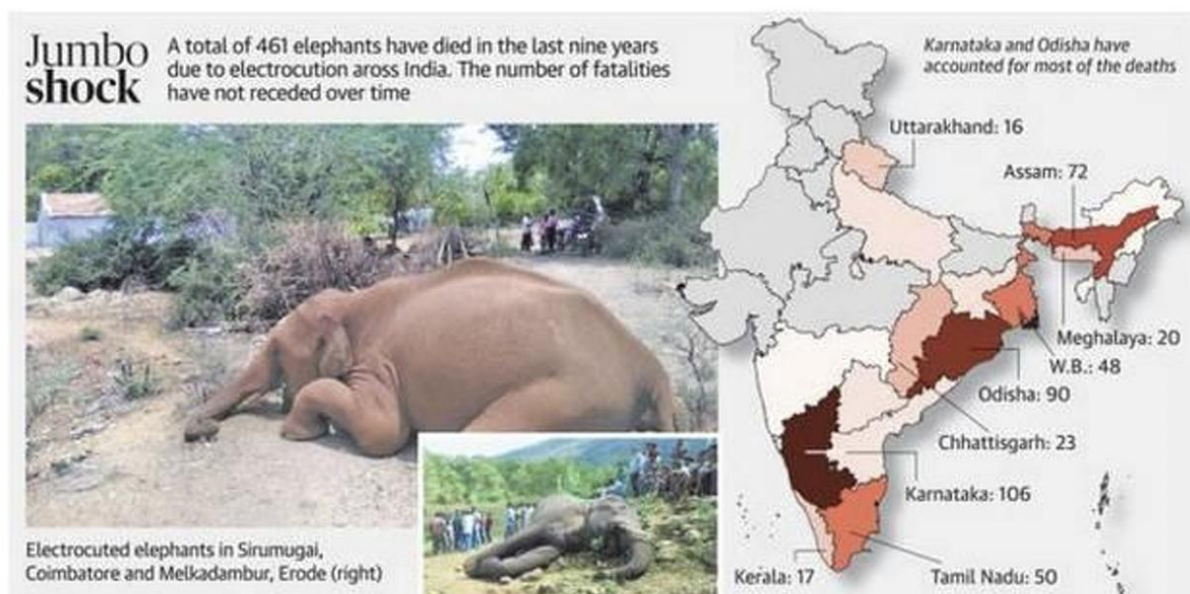


Figure 13: The image shows the state-wise statistics for the anthropogenic death of wild elephants in India between 2009-17. The figure is based on the data published by “The Hindu” newsletter in Nov, 2018 [21].

Illegal high-voltage fences are the main cause of indirect elephant deaths (Figures 9 & 13). As the deaths are concentrated in states of high elephant density (Karnataka, Assam), this establishes a direct argument that electric fences are a culprit for the conflict. Other lesser used fatal deterrents include the use of a poisoned bait, and retaliatory poaching [22].

6.2 . Proximity of Civilization to Elephant Habitats

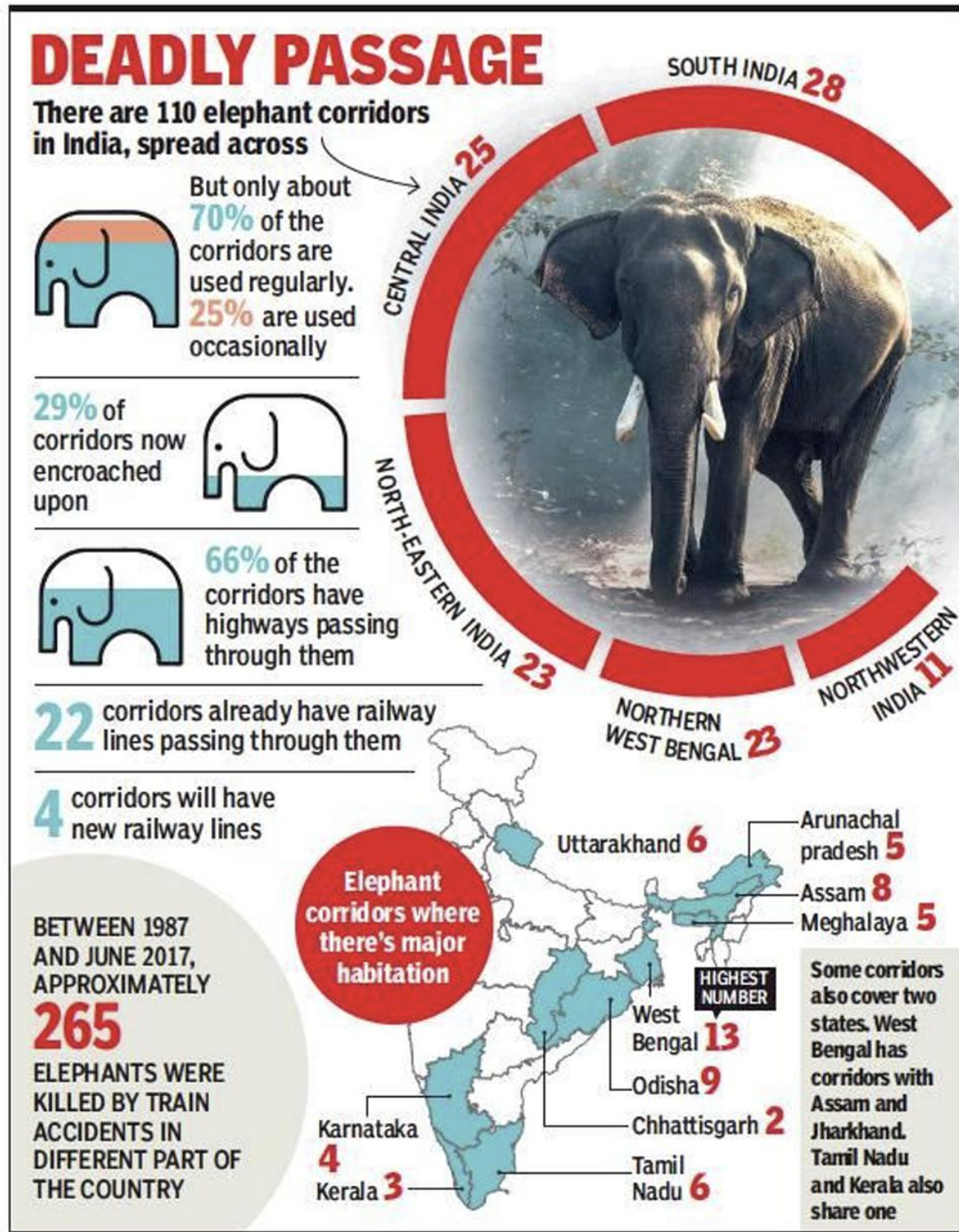


Figure 14: The figure shows the reasons for ineffectiveness of elephant corridors in the Indian landscape. The information about the corridors is as of 2018. The figure was published by "The Times of India" in Oct, 2018 [12].

Elephants entering forest-edge villages cause panic and trampling fatalities. They may also trample livestock and crops. Even basic settlements require infrastructure like farms, roads, and railways. Hence, the location of a settlement should be taken into consideration when planning a wildlife-corridor or elephant reserve to avoid potential causes of conflict(Figure-14).

6.3 . Ready Availability of Anthropogenic Resources

High-yield cropping-systems that lie near forest patches serve as a buffet invitation for elephants, creating the perfect conditions for conflict. It is documented that elephants prefer palatable, calorie-rich crops such as sugarcane and maize[4][14][23]. Insufficient livestock-grazing can also attract unwelcome wild herbivore guests.

Settlements near lakes or rivers are another conflict zone, especially during dry-seasons or if the nearby forest does not host a watering hole.

6.4. Natural and Ecological Variations: Dry seasons, Masting behaviour, drying grains

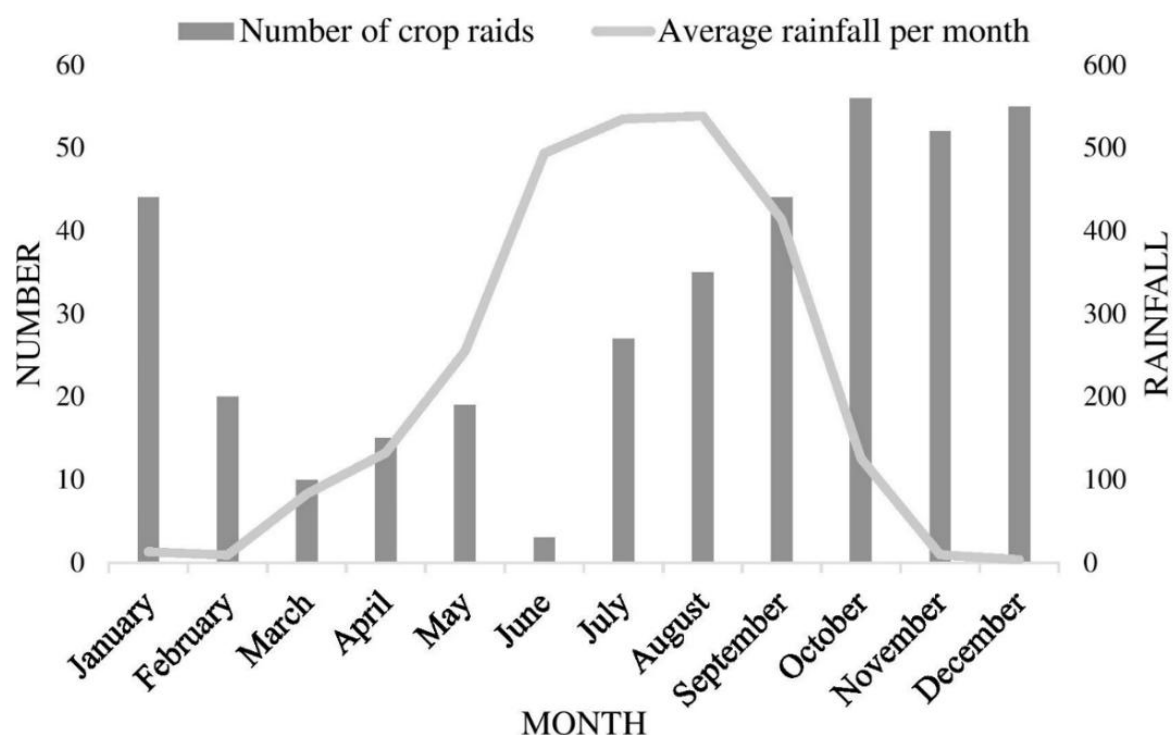


Figure 15: The graph shows the relationship between seasonal variation of precipitation and the crop-raiding frequency of the Indian Elephant (*Elephas maximus indicus*) between 2017-19 in North Bengal, India. The graph is sourced from Naha et al (2020) [23].

As evident from Figure-15, the crop-raiding frequency increases during the drier months, possibly due to lower foraging opportunities in the forest. This is also when most crops ripen,

showing that cropping-cycles induce strong seasonal foraging behaviour[24]. This effect can be amplified by climate-change as drought spells increase in some areas.

Elephants are also observed to more frequently raid fields during the night to avoid human activity. The raids are also dependant on the size of the herd. Mixed herds(containing calves) are likely to raid more intensively than lone bulls[14].

During musth (March–May), bulls become aggressive due to high testosterone. Other than the increased aggressive behaviour and expanded spatial ranges during this time(which multiplies chances of HEC), musth is a nutritionally heavy period, during which the lone tusker is more prone to raiding crop-fields[25].

6.5. Habitat Loss and Fragmentation



Figure 16: The figure shows the state-wise human death tolls caused by HECs between 2019-2024. The figure was published by “The Times of India” in Feb, 2025 using data from the Ministry of Environment, Forests and Climate Change (MoEFCC) [12].

An interesting visual correlation can be drawn where the states with the lowest elephant counts(Jharkhand, West-Bengal)(Figure-7) oddly enough, top the list where the most human fatalities have occurred(Figure-16). Along with Odisha and Karnataka(where the indirect HECs were maximum), these states have a relatively higher proportion of cultivated and managed areas within elephant habitats(Figures-6&8). Therefore, we can confidently say that the lack of dense, tree-covered habitat might be direct cause for crop-raids and the resultant HECs. The core forest areas in India have shrunk by ~40% between 1930-2013. This shrinkage in core and buffer zones drives elephants out to the farmlands[26].

It is hard to ignore the effects that ~1.4 billion human population can have on wildlife. As seen in Figure-14, Railroads, highways, and mining fragment habitats, worsening HECs. While the more direct threat of motor collisions is very real, the noise from the land transports also discourage elephants from using certain habitat corridors, reducing their spatial ranges and chances of migration(and therefore genetic intermixing)[2][25].

6.6. Regulatory Mismatches

Year	Human Deaths	Cattle loss, Crop damage etc. (No. of incidents)	Compensation paid (Rs lakh)
2019-20	102	6,063	714.2
2020-21	97	7,077	785.6
2021-22	152	7,094	1,056
2022-23	99	5,800	803.1
2023-24*	105	4,555	161.1
Total	555	30,589	3,519.90

Figure 17: The table shows the number of incidences of cattle and crop damage, and human casualties caused by the wild elephant raids, as well as the monetary compensation offered by the government per year. INR One Lakh ~ GBP 830. The figure was originally published by "The Times of India" in Feb, 2024 [12].

While it was seen in Chapter-4 that a comprehensive legal grid has been established to combat HEC, due to the massive human population and large spatial expanse, most of the frameworks lack central-enforcement, resulting in delegation of responsibilities to the state governments, which have conflicting agendas(such as land-use policies). Thus, decentralization of regulation ends up as the wound-up key for the conflict[2]. This problem is further catalysed by

insufficient compensations to the affected landowners for their monetary losses(Figure-17), a poor market access to especially rural areas, and blanket policies without regard for site-specific circumstances.

6.7. Literacy and Politico-Cultural Influence

Lastly, community literacy rates play an important role in reducing ignorance about effective strategies of dealing with HEC. Using Figure-18 as an example, learning to read/write/speak helps the locals access a wider range of literary sources, media, and technology, which creates a significant addition to the community's conflict resolution repertoire.

Furthermore, as seen in Section-5, local political and cultural backgrounds have an impactful, but not necessarily positive roles on how a community views and addresses HEC. Education provides exposure and trains an independent thought-process, which helps fight politico-cultural stereotypes that are misaligned with solving HECs.



Figure 18: The figure portrays how media can be used as a robust tool for spreading public awareness about avoiding HECs. The figure was originally published by "The Times of India" in Feb, 2025 [12].

7. Stakeholder Analysis and Power-Interest Relationships

7.1 . Stakeholder Identification and Categorization

“Elephants Should be Alive and Free to Roam!”	“Elephants Should be Confined in Numbers and/or Territory!”		“My Opinions Sway Depending on the Situation!”
	“Elephants are Trouble-Makers!”	“Elephants are Money-Makers!”	
Elephants	Land-Raid Affected Farmers and Property Owners	Ivory and Exotic Meat Poachers	International Regulatory and National Government Bodies
Religious Hindu Activists	Agricultural Market Drivers (Agricultural Product Consumers, Exporters, etc)	Elephant Product Consumers (Ivory, Meat, Dung, etc)	Political Activists
Conservation Activists	Human Rights Activists	Elephant Domesticators (Temples, Illegal Traders and Owners)	Scientific Community
Wildlife Tourism Industry	Logistical and Travel Benefitters (Rail and Roadway Users, Property Investors, etc)		Media
			Bystanding, Unaffected Local Population

7.2. Analysis of Stakeholder Interests

		Interests					
		Economic/ Consumer Benefits	Ethics / Public Image	Cultural History	Environment / Conservation	Scientific Research	Legal Compliance
S t a k e h o l d e r G r o u p s	Elephants	*	*	*	***	*	*
	Hindu Activists	*	**	***	*	*	**
	Conservation Activists	*	**	*	***	**	**
	Wildlife Tourism Industry	***	**	*	***	*	**
	Land-Raid Affected Farmers and Property Owners	***	*	*	*	*	**
	Agricultural Market Drivers	***	**	*	**	**	**
	Human Rights Activists	*	***	**	*	**	**
	Logistical and Travel Benefiters	***	*	*	*	*	**
	Poachers	***	*	*	*	*	*
	Elephant Product Consumers	***	*	*	*	*	*
	Elephant Domesticators	***	*	*	*	*	*
	(Inter)national Regulatory Bodies	*	***	**	**	**	***
	Political Activists	**	***	**	**	*	**
	Scientific Community	*	*	*	**	***	**
	Media	**	***	**	**	**	**
Bystanding Local Population	*	*	*	*	*	**	
	Total	32	28	22	27	23	29
Key: * = Less Important; ** = Important; *** = Very Important							

7.3. Power-Interest Grid

		Power		
		Low	Medium	High
Interest	Low	Non-Contributing Crowd: Bystanding Local Population	Contributing Crowd: Agricultural Market Drivers	
	Medium	Low-Stake Subjects: Logistical and Travel Benefitters, Elephant Domesticators	Passive Context Setters: Scientific Community, Elephant Product Consumers	Active Context Setters: (Inter)national Regulatory Bodies, Political Activists, Media
	High	High-Stake Subjects: Wildlife Tourism Industry	Passive Players: Hindu Activists, Conservation Activists, Human Rights Activists	Active Players: Elephants, Land-Raid Affected Farmers & Landowners, Poachers

7.4. Management of Stakeholders

		Power		
		Low	Medium	High
Interest	Low	Keep into account and Infrequently Monitor	Keep into account and Frequently Monitor	
	Medium	Keep Infrequently Informed	Frequently Monitor, Listen and Interact	Meet their Needs and Keep Satisfied
	High	Keep Frequently Informed and Suggest	Closely Manage	Constant Management

8. Recommendations

Recommendation	Method of Deployment	Rationale	Degree of Practical Success/Limitations	Examples
Land-Use Planning & Translocation	Moving an aggressive elephant	Minimize human-elephant interaction, but these methods should be used as a last resort	Translocation fails >80% as elephants return or die	Translocation attempts in Coimbatore[32]
	Relocating vulnerable villages		Uproots communities and violates civil rights	
	Designing and maintaining buffer zones			
Deterrent Crop-Planning and Diversification	Alternate cropping with deterrents(eg. citrus, chilli, onion, ginger, etc) on the fringes[28]	Deters elephants by smell and taste	Deterrent crops have lower value	Most common deterrent strategies in India
	Diversify with agroforestry or livestock-fodder plots		Needs farmer education in multiple agricultural disciplines	
Early Warning Technology	Install GPS collars on herd matriarchs	Alerts sent on smartphones and community radio	Trials show significant reduction in unexpected encounters[27]	Wildlife SOS's collar program in Chhattisgarh
	Drone, camera-trap & acoustic surveillance		GPS signal failures, Limited collars due to high cost	Not known
Community Vigilance	Patrols at night with megaphones	Drive elephants away, Warn other villagers	Fewer elephant raids	“Haathimitra Dal” in Chhattisgarh[14]
			Needs active community participation, and elephants adapt to human patterns[10]	Kruger-India collaborative experiments

Physical Barriers(Figure-19)[28]	Vegetation barriers	Deter elephants without entangling wildlife	Minimal upkeep	Eupatorium hedges
	Solar electric fences		60% drop in raid incidents	Tamil Nadu & Kerala
			Expensive installation, high maintenance, elephants learn to evade them[30]	
			Chilli-soaked rope/brick fences	Quick to learn and assemble
	Beehive fences		Noticeably lower raid incidents + income from honey	Original African model trialed in Tamil Nadu
			Bee-stings, hive upkeep	
Non-Lethal In situ Conflict Deterrents	Trip-alarms	Low-cost ways to scare-off/intimidate elephants	Elephants become habituated over time	Common methods
	Waving light Torches			
	Firecrackers			
	Burning chilli smoke			
Enhanced Compensation and Insurance Schemes[15]	INR = GBP 8350 for every human fatality & INR = GBP 1650 for grievous injuries from elephant attacks	National compensation is an increment in value from the previous scheme while crop-damage compensations are provided by individual state govts on a per-acre damage norm. It buys conflict tolerance.	Too early to comment but should be relatively successful	Nation-wide scheme implemented by MoEFCC
			Heavy on govt pockets	
	Crop-insurance scheme		Claims are tedious for farmers	Pilots in Mizoram
			Can be a moral hazard if farmers start assuming the loss is insured	
Tightened Legal Measures	Stricter fines on AC electric fences	Avoid lethal conflict for elephants	Legislation not yet passed	
	Regulated train speeds through forested zones		Unknown	Kerala has posted speed-limit signs[15]

Corridor Management and Governance	Eco-sensitive zoning	Securing identified elephant corridors	Successful management till now	Assam demolished illegal boundary wall along elephant corridor[30]
	Purchase of corridor land by govt		Limits economic development and can be politically unpopular	Ficus and Bamboo plantations in Odisha[30]
	Restoring degraded patches			
	Restore forest watering holes			
Improve Inter-agency Government Collaboration	Project Elephant collaboration with “Compensatory Afforestation Fund or MNREGA program	Increases efficiency by facilitating fund allocation and man-power from other departments	Not yet done	Cooperation of MoEFCC with Indian Railways[15]
			Time-consuming legal process	
Improve Literacy & Public Awareness	HEC awareness drives	Communities get knowledge on handling HECs	Moderate success but language barriers and cultural beliefs are hard to penetrate	Elephant Family Foundation India conducts education drive
	Literacy drives			Total Literacy Campaign, Digital India initiative

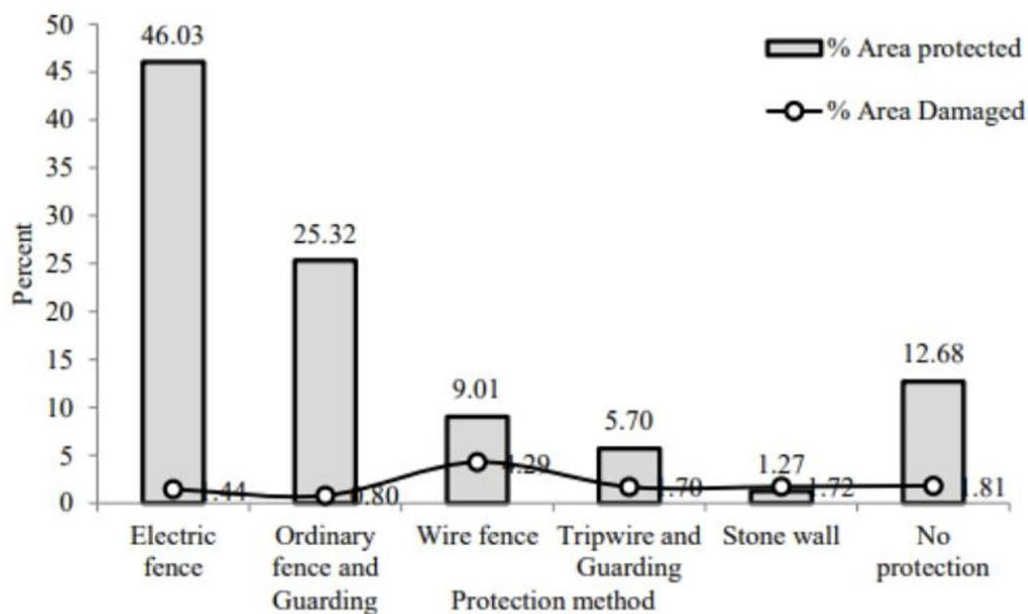


Figure 19: The bar chart shows the different methods used by the villagers to protect their crops and their efficacy during elephant raids. The graph is copyrighted by the Wildlife SOS organisation [33].

9. Conclusion

Human–elephant conflict(HEC) in India epitomizes the tension between biodiversity conservation and rural livelihoods. Despite India hosting the world’s largest Asian Elephant population, habitat fragmentation, expanding agriculture, and unsafe deterrents such as electric fences continue to escalate risks for both people and elephants. The persistence of crop-raiding, fatalities, and property loss underscores the inadequacy of piecemeal solutions and the need for integrated strategies.

Policy frameworks like the Wildlife Protection Act(1972) and Project Elephant have laid a strong foundation, yet enforcement gaps and uneven compensation schemes weaken their impact. Community-based approaches, including deterrent crops, beehive fences, and vigilance patrols, show promise but require sustained participation and education. Technological innovations such as GPS collars and acoustic monitoring further demonstrate how science can reduce unexpected encounters.

Cultural reverence for elephants, embodied in traditions surrounding Ganesha and temple ceremonies, offers both opportunities and challenges for conflict mitigation. Harnessing this cultural capital alongside ecological planning can foster tolerance and coexistence. Ultimately, securing elephant corridors, restoring degraded habitats, and improving literacy and awareness are essential steps toward balancing conservation with human needs. Only through coordinated governance, community empowerment, and adaptive management can India safeguard its heritage species while protecting rural livelihoods.

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11. Appendix 1: Gen-AI Coversheet

Completion and submission of this coversheet is mandatory for Category 2 and Category 3 assessments. Additional cover-sheets/declarations relating to Academic Integrity may also be used.



Gen-AI in assessment declaration

Student 500 number: 500761446

Module Code: ENS-4211

Assessment: Critical_Report

Assessment's Gen-AI Category (see the Assessment Brief): 2

The following statement reflects my use of Gen-AI within this assessment (select all that apply):

Description of use	Select (X)
Gen-AI has not been used at all in producing this assessment.	
Gen-AI was used in an assistive role to research topics and provide explanations.	X
Gen-AI was used in an assistive role to suggest sources of information, e.g. publications to read.	X
Gen-AI was used in an assistive role to summarize the content of publications or other sources.	X
Gen-AI was used in an assistive role to produce an outline or draft structure of my assessment. Note: for Category 2 assessments, the fundamental content of the submission (e.g., the arguments, interpretations, conclusions etc) must be your own.	X
Gen-AI was used in an assistive role to improve spelling, grammar, sentence structure and/or clarity of expression of my assessment. Note: for Category 2 assessments, the basic written expression must be your own. Therefore whilst Gen-AI may correct mistakes it must not produce the actual content.	X

Gen-AI was used in an assistive role to translate text during the production of the work.	
Gen-AI was used in an assistive role to analyse data and/or produce visualizations of data.	
Gen-AI was used in an assistive role to help produce creative outputs e.g. art, imagery, movie, music, creative writing.	
Gen-AI was used in an assistive role to refine computer code and/or other output.	
Gen-AI was used in another assistive role (please describe):	
Gen-AI was fully used to complete a Category 3 assessment and use complied with the assessment instructions provided.	

By using this cover sheet and entering my student ID above, I declare that I have provided honest and accurate details regarding my use of Gen-AI within this assessment.